Remedy ledger

What was tried against the block-1 pathologies, what it bought in pathology counts and in scores, and what remains untried

Technical note

31 July 2026

Abstract

Companion to `tex/ar2_pathology_ledger`. Two remedies were implemented and measured on the same 40 cells with identical seeds: assertion-guarded reseeding (pipeline level) and the sign-anchored prior $\lambda \sim \mathrm{HN}(0, 20)$ (model level). Both remove the score-relevant pathology; their delivered scores are statistically indistinguishable (paired over 35 cells, $p = 0.52$ CRPS, $p = 0.28$ Brier), and the study's headline conclusion survives both, sharpening under the prior. The ledger closes with the untried remedies and the argument for why their expected score gain is bounded near zero.

1 Remedy 1 (pipeline): assertion-guarded reseeding

Pre-registered rule, symmetric across methods. A fit failing truth recovery (B) or predictive spread (C, $\max_h \mathrm{sd}(\hat{y}_h) \leq 3\times$ marginal SD) is refitted with seed $+ 7919 \cdot$ attempt, at most three reseeds; a cell still failing is kept but flagged and excluded from the primary analysis. Assertions use only the fit, the training data, and the simulation truth — never validation outcomes.

Table 1: What the guard bought (bit-for-bit counterfactual of the unguarded study; leads 1–5).

	unguarded (att-0)	guarded (kept)
failing cells	13/40	3/40 flagged
CRPS, affected cells	2.30	2.03 (−12%)
Brier q_{95}, affected cells	.0437	.0368 (−16%)
largest single-cell rescue	Brier .0205 → .0017 (12×)	
cost	≈ 1.33 fits per affected cell	

The headline comparison was additionally shown to be insensitive to the flag exclusions (sensitivity run keeping all cells, $p = 0.010$–0.021).

2 Remedy 2 (model): $\lambda \sim \mathrm{HN}(0, 20)$

Backend flag `lambda.positive` (default off; production behavior unchanged). The joint (β, λ) full conditional is multivariate normal; under the truncated prior the exact Gibbs draw is

$$\lambda \mid \cdot \sim \mathrm{TN}_{(0,\infty)}\big(m_\lambda, V_{\lambda\lambda}\big), \qquad \beta \mid \lambda, \cdot \sim N\big(m_\beta + V_{\beta\lambda} V_{\lambda\lambda}^{-1}(\lambda - m_\lambda),\ V_{\beta\beta} - V_{\beta\lambda} V_{\lambda\lambda}^{-1} V_{\lambda\beta}\big), \quad (1)$$

Table 2: The HN experiment. Same 40 cells, same attempt-0 seeds, single fit per cell, no guard, no reseeding.

	$N(0, 20)$ att-0	HN$(0, 20)$
failing cells	13/40	4/40
reflections ($\hat{\lambda} < 0$)	10	0 (by construction)
previously unfixable flag healed	—	s5 d8 m2: $\hat{\lambda} = +3.63$
healthy cells harmed	—	0/27
parameter side effects, paired over 35 mutually clean cells:		
β_0		+0.020 ($p = .042$; 0.2% of truth)
$\lambda,\ a,\ b,\ \phi_z,\ \phi_\tau$		all $p > 0.19$
scores vs the guarded study, paired over the same 35 cells:		
CRPS		2.165 vs 2.166 ($p = 0.52$)
Brier		.0491 vs .0494 ($p = 0.28$)

with the deep-tail truncated draw by exponential rejection (Robert, 1995). No approximation is involved; the reflected branch has prior mass zero and disappears from the posterior.

The four residual failures are, by pathology, U×2 (s5 d2, both methods; no prior can supply identification the window lacks), P (s5 d3 m2, $\hat{\lambda} = 4.97$; the prior constrains the sign, not the scale), and H (s4 d6 m2, the spread-statistic coin flip of the pathology ledger, §4).

Effect on the headline. Under the guarded study the AR(2) advantage at AR(2)$(0.15, 0.80)$ was CRPS -31.3% ($p = .039$); under the HN study it is -35.4% ($p = .012$). The conclusion does not depend on which remedy protects the study.

Adopted policy (31 July 2026). From Experiment A onward the production pipeline uses the HN prior alone. Assertion C is demoted to a descriptive record — no fit-level gate, no reseeding, no cell exclusion — so the primary analysis consumes all cells. The justification is Table 2: the prior alone delivers guard-equivalent scores at one fit per cell, and the pathologies it leaves (U, P, H) have no measured score path.

3 Untried remedies and their expected value

The score gain of any further sampler- or prior-side remedy is bounded by an equivalence argument. Two independent mechanisms — selection by reseeding and removal by truncation — already agree on the delivered scores to within noise ($p = 0.52/0.28$); a third mechanism that removes the same pathology can only reproduce the same healthy posterior, so its expected score gain over the adopted policy is ≈ 0. The value of the entries below is therefore robustness or diagnostics, not score.

4 What remains unprotected, quantified

Under the adopted HN-only policy the primary analysis includes the four residual cells. Their measured score footprint, leads 1–5:

Table 3: Untried remedies. Expected gains are predictions, not measurements.

remedy	targets	expected pathology gain	expected score gain / cost
Mode-jumping swap kernel: deterministic proposal $(\lambda, \beta_0, z) \to (-\lambda, \beta_0', \text{flipped } z)$ with MH correction	R	same as HN (mixes across branches; post-hoc relabel)	≈ 0; high engineering cost, sampler revalidation
Tighter `lambda.s` (e.g. scale so $\lvert\lambda\rvert = 12$ costs $\gg$ a few nats)	R, P	partial; starves both branches' tails but shrinks honest estimates	≤ 0 (bias risk); scale choice is a judgment call
Quantile-spread assertion, $q_{99} - q_1$ in place of $\max_h$ sd	H (false alarms)	removes the $a < 2$ instability of the C statistic entirely (finite for all $a > 0$)	0 (diagnostic only); trivial cost; moot while C is descriptive
Longer training window	U, τ-shape recovery	raises effective sample size of the z path and the τ marginal	unknown for scores; changes the study design (no longer 50-day short-window question)
Informative (a, b) prior anchored at truth	H, τ recovery	prevents $\hat{a} < 2$ realizations	≈ 0 for paired scores (both methods share the cell's τ); indefensible outside simulation

- **U (s5 d2, both methods).** CRPS 1.164 vs 1.158, Brier .0194 vs .0190 (i.i.d. vs AR(2)) — the two methods agree on this dataset to three decimals, so the paired difference it contributes is ≈ 0. Including it dilutes, never distorts.
- **P (s5 d3 m2).** The inflated fit scores *better* than its healthy replacement (CRPS 1.99 vs 2.15), and d3 is an AR(2)-losing dataset under either version; inclusion is conservative for the headline.
- **H (s4 d6 m2).** Predictive cloud indistinguishable from the passing chains to $q_{99.9}$; scores within 1% of the guarded version (.1338 vs .1349). No footprint.